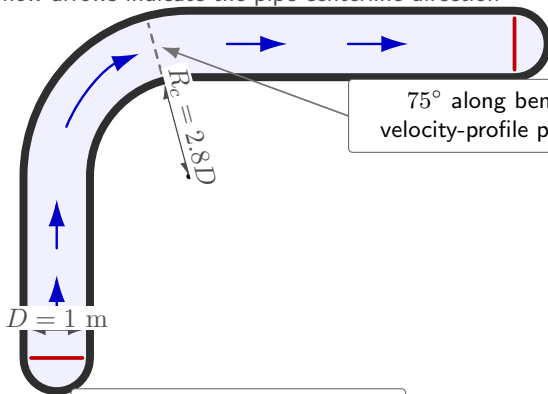


VMFL030: 3D 90° pipe bend

flow arrows indicate the pipe centerline direction

pressure outlet
 $p = 0 \text{ Pa}$

75° along bend
velocity-profile plane



fully developed
turbulent inlet profile
zero transverse velocity

shown symmetry plane
3D half-domain simulation
 $Re_D \approx 43,000$